



Drill bit strengthened with the alloy tungsten carbide

*This mineral is the main ore of tungsten.*



Wolframite

## Tungsten

74  
**W**

Heavy and dense, tungsten has the highest melting point of any metal, turning to liquid at 6,177.2°F (3,414°C). It was once used in light bulb filaments, but these kinds of bulbs have now been phased out because they are inefficient.

## Osmium

Fountain pen

76  
**Os**

The pure form of osmium forms a poisonous strong-smelling oxide when it reacts with oxygen in the air. For safe use, it is combined with other elements. From fingerprint powder used in forensic laboratories to nibs for fountain pens, osmium alloys have many specialized uses.

*The nib of this fountain pen is made of an osmium alloy, making it hard-wearing.*



Osmiridium sand

## Rhenium

75  
**Re**

Named after the Rhine River in Germany, this element was discovered in 1925. Rhenium has the second-highest melting point of all metals. Its ability to withstand high heat has made it very effective in superalloys used in oven filaments, X-ray machines, and jet aircraft turbine blades.



Pure rhenium pellet refined in a laboratory

### Turbo engine

The Rolls-Royce Trent XWB is a turbofan jet engine used on the Airbus A350 XWB airplane. Its inner blades, which are made of a superalloy that includes rhenium, can withstand extremely high temperatures.



## Iridium

77  
**Ir**

In nature, iridium occurs as an impurity in naturally occurring platinum. A thin layer of iridium is also present in Earth's crust. Some scientists believe that the meteorite that triggered the death of the dinosaurs 66 million years ago also distributed this layer of iridium across Earth with its impact.

### Powerful telescope

NASA's Chandra X-ray Observatory, an Earth-orbiting telescope that studies X-rays from distant stars, has eight iridium-coated glass mirrors. The coating helps catch and bounce the X-rays toward the telescope's scientific instruments that study the rays and gather data.

Artist's impression of Chandra X-ray Observatory

